

Global Aerospace & Defense

Space Exploration Technologies Corporation

Rating

Outperform

Price Target

SPCX

239.00 USD



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SpaceX: The debates - Investor views and controversies as we head toward the first earnings report

Since SpaceX's IPO on 12 June 2026, we have been on the road speaking with investors about the stock's long-term opportunity, following our recent initiation, "[Space Exploration Technologies Corporation: Doing Time in the Universal Mind - Initiate at Outperform; Target Price \\$239](#)". At the same time, we have seen high volatility in the stock, with a sharp decline since its peak after the IPO. Here, we discuss the questions raised by investors regarding the fundamentals for the stock - fundamentals centered on the ability of SpaceX to launch thousands of satellites into orbit to form a data center constellation.

There are four topics that we see as central to the success of the SpaceX valuation. Most important is achieving rapid reuse of Starship. The other three are about having sufficient semiconductor capacity, moving through regulatory processes, and that the need for compute capacity to remain high. Three topics that are legitimate areas of debate, but less relevant to the value proposition of SpaceX are China as a competitive threat, attractiveness of global Starlink broadband, and the feasibility of the company's direct to device mobile plan. On this last topic, we also have skepticism regarding the potential of mobile. Lastly, we highlight four topics where we have seen pushback on the technical feasibility of orbital data centers - none of which we see as material issues. Those are about the ability to cool data center satellites, the source of solar energy, avoiding damage from micro-meteorite strikes, and latency related to space-based data centers.

SpaceX will report its first set of results on August 4th after the market close. We believe the quarterly results should not matter. What will be important is the level of confidence projected by management regarding the company's growth path. Investors should look beyond short term stock movements as we view the case for a multi-trillion dollar valuation is about "if" not "when", for orbital data center plans.

Investment Implications

We rate SPCX Outperform, TP \$239, using sum of the parts on 2031 EBITDA for each segment, assuming a one-year delay on orbital data center launches. We use late stage venture discount rates, 25% and 35%, respectively, for Connectivity and AI.

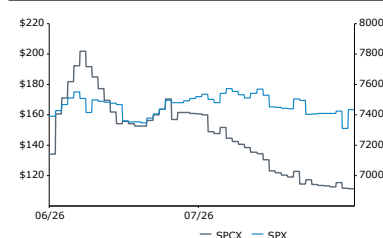
Adjusted EPS	F25A	F26E	F27E	Financials	F25A	F26E	F27E	CAGR	Valuation Metrics	F25A	F26E	F27E
SPCX (USD)	--	(0.18)	0.83	Revenues (M)	18,674	40,236	84,990	--	Adjusted P/E (x)	(608.1)	136.0	
				Adjusted EBITDA	6,584.0	17,131.0	43,447.0	--	EV/Adj EBITDA (x)	226.3	87.0	34.3

Source: Bloomberg, Bernstein estimates and analysis.

Close Date	30 Jul 2026			
SPCX Close Price (USD)	112.20			
Price Target (USD)	239.00			
Upside/(Downside)	113%			
52-Week Range	225.64/107.01			
SPX	7,437.63			
FYE	Dec			
Div Yield	NA			
Market Cap (USD) (B)	1,476.46			
EV (USD) (B)	1,490.10			
Performance	YTD	1M	6M	12M
Absolute (%)	NA	(34.3)	NA	NA
SPX (%)	8.6	(0.8)	7.2	16.9
Relative (%)	NA	(33.5)	NA	NA

Source: Bloomberg, Bernstein estimates and analysis.

Price Performance, 1YR



DETAILS

Following our initiation investors have challenged us on almost every aspect of the story – from the feasibility of launching from multiple sites across the US, to the practicality of cooling orbital data centers. In this note, we distill 10 key insights from those client conversation.

THE TOPICS THAT MATTER MOST - MAKE OR BREAK ON THE GROWTH MODEL

1. Reusability of Starship rockets - The most critical issue - Will the heat shield hold up?

We see the key issue for the SpaceX value proposition as reuse of the Starship rocket. This topic has been an important issue for many investors, as we believe it should be. Our model has roughly 3,600 Starship launches in 2031 (the company's expectations are much higher), which can only be done with a fully reusable rocket (both stages). So far, the company's expectations has landed Starship V2 boosters on the launchpad, but not yet V3 boosters. Launch 13, which was completed on July 20th, was largely successful, although some engines failed to light on the booster ahead of splashdown. Our assumption is that booster landings with reuse will be achieved as it was with the V2 rockets.

The important success on Launch 13 was the successful recovery of the Starship itself (second stage) with the heat shield much more intact than prior launches. We await the company's assessment of the damage to the Starship heat shield. If there is minimal damage, then the company is moving close to reuse. After Launch 13, Elon Musk said that SpaceX may attempt to do a "chopsticks" catch of the second stage on Launch 14. Demonstrating catches of the booster (Super Heavy) and then the second stage (Starship) is the first step toward the critical achievement of rapid reuse. So far, we see the progress as good. But, there is a long way to go. Reaching our launch goal will still require at least daily launches. The frequency of launches will also be tied to the number of launchpads. Today there are two, with the addition of two more coming. The company is in negotiations with state governments to determine where the next five or six will be located.

2. Semiconductor availability - Will this limit growth?

The required semiconductor capacity for SpaceX's orbital data center buildout has been another area of investor pushback. We view this as a valid concern, but still see a credible path for SpaceX to secure sufficient semiconductor supply over time.

Under the assumption that a data center satellite will use a similar architecture, we can estimate how much semiconductor manufacturing capacity will be needed by the company. And the numbers are not small. Our model calls for ~20 GW of incremental orbital capacity to be deployed in 2031, which would require close to 170K satellites to be launched assuming ~120 kW per satellite (the current spec of the AI1), requiring ~2.8 million wafers of annual production capacity. At a fab size of ~50,000 wafer starts per month this would translate to ~5 dedicated semiconductor fabs solely serving SpaceX, probably costing somewhere north of ~\$160B to put in place over the next several years (Exhibit 1). This is a large number, though not necessarily impossible (especially as some will still likely be third-party) and compares reasonably favorably with current company Terafab investment plans (~\$119B forecast as of this writing) though the timeframe to bring this amount of capacity online is ambitious especially as SpaceX has not done this before. If they cannot deliver on time it will put more pressure on them to secure large amounts of incremental capacity from existing foundry and memory players.

EXHIBIT 1: Our 2031 model would likely require ~5 dedicated semiconductor fabs serving only SpaceX

Deployed Orbital Capacity (GW/Year)	Thousands of Satellites Required @120 kW per Satellite	Millions of Wafers Required per Year		Cost @\$35B per Fab (\$ Billions)
		@17 Wafers per Satellite	# New Semiconductor Fabs Required @50K WSPM per Fab	
5	42	0.7	1.2	\$41
10	83	1.4	2.4	\$83
15	125	2.1	3.5	\$124
20	167	2.8	4.7	\$165
25	208	3.5	5.9	\$207
30	250	4.3	7.1	\$248
35	292	5.0	8.3	\$289
40	333	5.7	9.5	\$331
45	375	6.4	10.6	\$372
50	417	7.1	11.8	\$414

Source: Company reports, Bernstein estimates and analysis

3. Will regulatory considerations be a problem?

We have heard a number of investors express skepticism about being able to accelerate the SpaceX vision and remain in compliance with regulatory authorities. The FAA must approve the flight path for each Starship launch. So far, the FAA has not cleared SpaceX for an orbital Starship flight. The hope is that Launch 14 may be able to go orbital. But, the FAA performs environmental assessments that consider the impact on airline flight paths, as well as potential environmental damage and risks to wildlife, including sea life. The current Environmental Assessment has many strict considerations, with August 3rd as the deadline for public input. But, this week the FAA announced that it would relax environmental restrictions because of the importance of enabling a higher cadence of space launches. The assessment related to air traffic would remain. While changes made should enable more Starship launches, it is not yet clear how the regulatory path will go when thousands of launches are under consideration and a potentially less supportive administration could be in place.

The FCC must approve satellite communications programs. A fast track process was recently approved. But, this will not apply the very large constellations proposed by SpaceX. Recent filings include proposals for a ~100,000-satellite Starlink V3 constellation and, separately, a potential large-scale AI data center constellation (up to ~1 million units), neither of which has yet received full regulatory approval. Instead, SpaceX is also looking at incremental steps to build up Starlink's capacity.

There are many other regulatory questions that will need to be resolved. Each country will have its own regulatory environment for telecommunications. The issue of data sovereignty is another important topic. Many countries have rules that prevent data from being stored in other countries. But, the argument for orbital data centers is that data stored in space is not resident in any country and consequently has no issues regarding data sovereignty. But, it is not yet clear exactly how space as a location will ultimately be treated.

4. Volume of compute needed - Is AI compute a bubble?

The need for orbital data centers starts with the assumption that there is a massive need for compute capability. Elon Musk has suggested that SpaceX would eventually be launching 1 terawatt of compute into space each year. The plans we are working with now are far more modest. But, making the orbital data center program a success still depends on having a huge volume of compute power. To this day, we have never seen the world have excess compute. Possibly, the rush to add capacity to fuel AI models is too rapid. But, even if that is the case, there should still be value in having a lower cost solution for the compute capacity that is needed, relative to terrestrial capacity.

LEGITIMATE QUESTIONS - ALTHOUGH THEY DO NOT CHANGE THE ULTIMATE VALUE PROPOSITION

5. Rising Chinese Competition

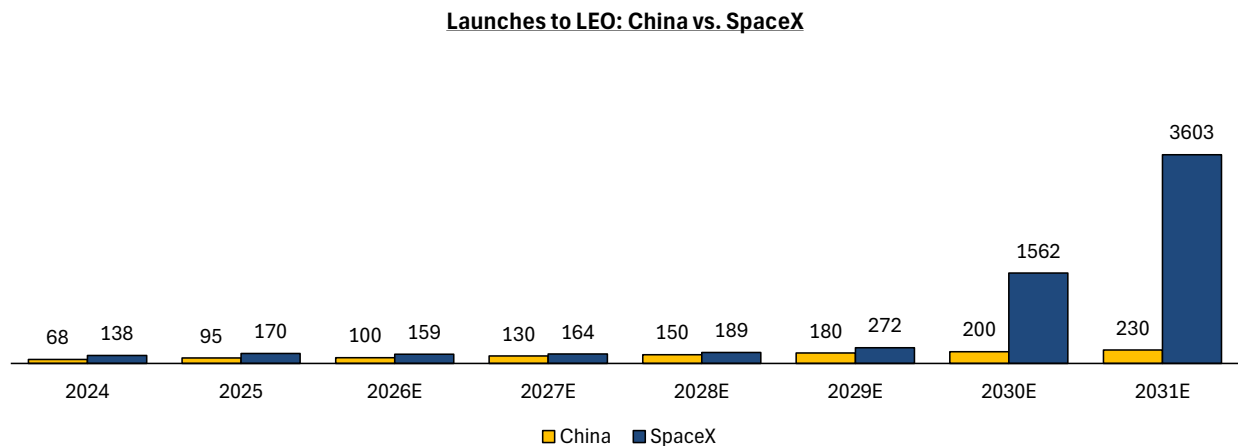
In our discussions with SpaceX management, China was identified as the most serious competitor. The country has tremendous resources that it is willing to use for achieving leadership in space. Nevertheless, China is still far behind SpaceX.

On July 10th, China successfully landed the first-stage booster from a Long March 10B rocket launch. Since then, a lot of investors have questioned us around this rising competition and this remake of the Space race. We see China as the lead competitor for SpaceX. This launch and recovery comes roughly six months earlier than we expected. China is moving aggressively to place a research station on the moon and create large LEO satellite constellations. In December, China filed with the ITU to place more than 200,000 satellites in low earth orbit. Despite China's success in landing its Long March 10B, it is far behind SpaceX. This was the first successful landing of a large launch vehicle booster. It has yet to do the reuse of a booster. SpaceX has been relaunching boosters for nearly ten years with its Falcon 9 rocket. 165 launches were done last year, which also required a robust production line for second stages. To reach a high volume of launch, China will need to demonstrate relaunch capabilities and move to high rate production. Most importantly, Long March 10 is only first stage reuse, whereas Starship is intended to be full reuse (not yet demonstrated).

A key part of the Long March program is landing humans on the moon (official target 2030), leading to an eventual lunar research station. SpaceX is a major part of NASA's space program, which has a goal of a manned moon landing by the end of 2028. There is an emerging space race here, which will likely increase U.S. government support for SpaceX (along with others, such as Blue Origin and Boeing).

As discussed in our SpaceX initiation, we see no real competition in the near-term for SpaceX on either launch capability or satellite operations. But, given China's massive resources and long-standing goals, China must be considered a serious competitor over the long-term. It still appears to us that China lags SpaceX by something on the order of ten years, assuming SpaceX is successful in achieving its goals.

EXHIBIT 2: **Even if China manages to reach its ambitious targets (which we view as very unlikely) it would still meaningfully lag SpaceX**



The forward targets for China represent aspirational government-backed projections
Source: PayloadSpace, Company report, AEI Space, Bernstein analysis and estimates

6. Starlink broadband penetration - Is there an issue here?

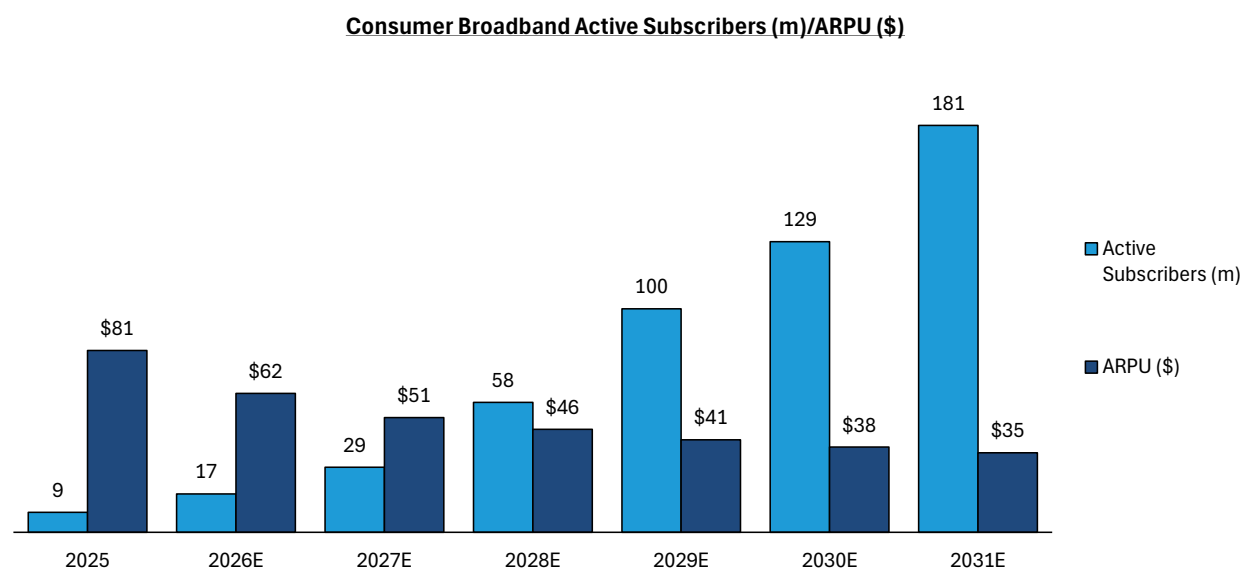
We are expecting rapid growth in Starlink's broadband businesses for consumer, enterprise, and government. The main source of pushback here has been among telecom-oriented investors who see terrestrial fiber as a better option for all but rural users. The most powerful application will be with users in locations that are not served by cable or fiber (where the Starlink share will likely be 100%). But, we expect there will be much broader adoption by suburban users. Even though there is more latency with Starlink (roughly 23ms round trip), this is not a problem even for most gaming applications. Surveys show that Starlink has the highest customer satisfaction rating of any consumer broadband service. In addition, Starlink has the advantage of having an inherently global network so that it can add subscribers in any country with essentially zero additional cost. This gives Starlink

exceptional flexibility in pricing. With its current 9,600 satellite constellation and planned capacity expansion by more than a factor of ten, no other provider is remotely close. Our forecast for global subscribers and ARPU is shown in Exhibit 3.

Starlink also has the advantage of supporting an enterprise and government customer base, which further spreads cost. The U.S. government invests in the defense version of Starlink (Starshield), while many other countries are adopting Starlink for monitor of remote regions, emergency services, and border control (e.g., UK, Australia, Ukraine, Japan, Netherlands, Poland). On the enterprise side, Starlink offers a global reach that other services lack, in addition to mobile applications where no competitor comes close to the performance (e.g., maritime, aviation). For broadband services, we expect high growth from this business that is already highly profitable.

Risks to US telecom providers from Starlink were highlighted by Bernstein's US Telecom team: [US Telecom: Adjusting price targets reflecting Starlink risks](#) (which resulted in a material PT cut in response to the Starlink threat).

EXHIBIT 3: Starlink's ARPU goes down over time as it intensifies its international expansion



Source: Company reports, Bernstein analysis and estimates

7. Can direct-to-device work for Starlink? The path is not clear

An area where we have heard strong pushback on SpaceX plans is for direct-to-device mobile service. On this one, we agree. Our outlook for the D2D business is far more conservative than what SpaceX has described. In terms of SpaceX as a stock, however, success will not depend on this business. In the end, success will depend primarily on orbital data centers, where the issues at the beginning of this report are most relevant.

SpaceX currently has a relatively small network of V1 Mobile satellites that primarily provide deadzone coverage for mobile network operators (MNOs). The company's plans are to expand this network with its coming V2 satellites. These satellites should each have roughly 16x the capacity of the V1 satellites, with higher power. This new V2 mobile network should begin launches in 2027.

Many telecom-oriented investors see success here as highly unlikely due to the technical limitations of the satellite network and the need for a terrestrial capability to complement the satellite constellation. The company is targeting deals with MNOs to effectively become an MVNO (like Charter, Comcast have done). So far, there is strong resistance in the U.S., although it is very possible that one player (i.e., T-Mobile, AT&T, Verizon) might break and link up with Starlink. But, all this is uncertain. In the absence of a partner, SpaceX has said that it could build its own terrestrial network - something that we find daunting.

In any case, we agree with skeptical investors on this topic. It remains to be seen how this might play out. In the end, however, we believe that success in this business is tangential to the overall case for SpaceX.

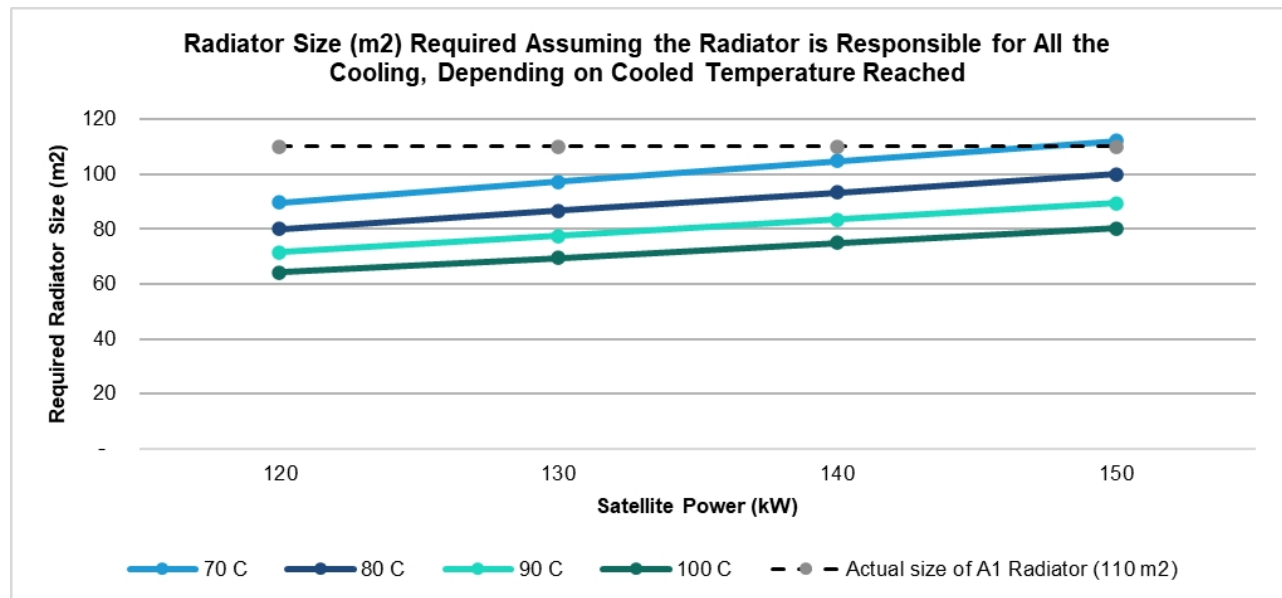
TECHNICAL ISSUES WE BELIEVE ARE RESOLVED - BUT, THESE QUESTIONS KEEP COMING UP

8..Cooling datacenters - This is not a problem!

Questions about pace-based data center cooling continue to come up in our investor discussions. First, it would be surprising if this issue had somehow been overlooked by the engineers at SpaceX. Nevertheless, we addressed this directly in our initiation report by explicitly modeling the relevant heat-transfer dynamics. In short, we conclude that cooling data center satellites in orbit is technically 100% viable.

Given that both senior analysts, before spending two decades in the securities industry, were Ph.D. physicists, we decided to do our own analysis. To assess feasibility, we analyzed the thermal environment of a representative data center satellite operating in sun-synchronous orbit, where continuous solar exposure enables stable power generation while dedicated radiators dissipate waste heat. Applying standard heat-transfer principles, we modeled the equilibrium between onboard power generation, absorbed heat, and thermal radiation. In the vacuum of space, heat rejection occurs via radiation rather than convection, making cooling capacity primarily a function of radiator surface area and operating temperature. Our analysis indicates that, with the inclusion of dedicated radiators, satellites supporting in excess of 100 kW of compute load can maintain acceptable thermal conditions while remaining within plausible launch constraints. While meaningful engineering and execution challenges remain, the governing physics do not represent a fundamental barrier. We therefore view in-orbit cooling of high-power AI data center satellites as feasible. A detailed discussion of the heat-transfer physics, assumptions, derivations, and sensitivity analyses is provided in the Appendix.

EXHIBIT 4: **Our math suggests a radiator size requirement very consistent with what the company is proposing for their AI1 satellite**

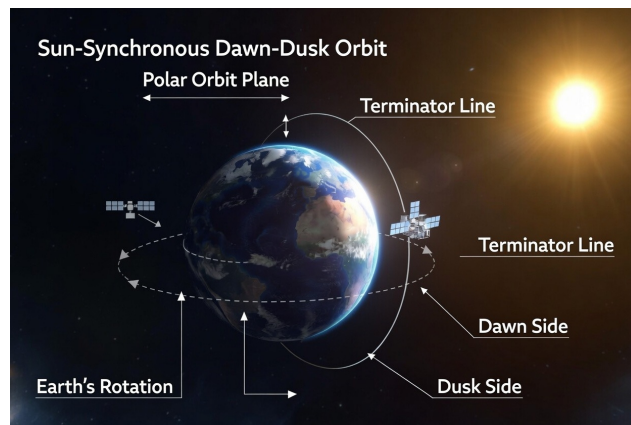


Source: Company reports, Bernstein analysis

9. Orbital Datacenters' sun exposure - Yes, it is always on

Concerns have been raised regarding the primary power source for orbital data centers—solar energy—and the assumption that satellites would only be exposed to sun roughly half of the time. This assumption is wrong. Orbital data centers would be deployed in a sun-synchronous dawn–dusk orbit, which allows the satellite to remain aligned with the terminator line and maintain near-continuous solar exposure (Exhibit 5). As SpaceX says: “It’s always sunny in Space”.

EXHIBIT 5: A sun-synchronous dawn-dusk orbit hugs the terminator line, keeping the satellite in the sun



Source: Grok

10. Latency of orbital datacenters - Not an issue for most inference applications

The latency surrounding orbital data centers has been put in question by investors multiple times. However, laser communications have already been demonstrated on Starlink, which is now a 9,600 satellite network. We do not expect latency to be an issue, as the delay from a 600-800km high orbit is 2 milliseconds. This latency is less than that for users of many terrestrial data centers.

A related question has been about latency within a constellation across many orbital data centers. If one requires a large cohesive data center for training, the terrestrial option is likely better. The intent at SpaceX is to perform most training activity at its terrestrial sites (e.g., MACROHARD) as it does today. Inference would be done with orbital data centers. Because demand over time for compute is expected to move heavily toward inference, the orbital data centers should provide the lowest cost solution for this largest portion of demand. There are applications that may be better suited for collocation operations on earth, such as high-frequency trading. But, such applications are expected to be a small part of total demand.

11. Micrometeorite strikes and space debris - A real problem, but we believe it is understood

We have had investor questions about space debris and micrometeorites strikes hitting the large solar panels and radiators on data center satellites. This is a serious consideration for any orbiting spacecraft. It is an issue that satellite operators have dealt with for many years, including with the International Space Station. SpaceX has been addressing this on its Starlink satellites by sensing objects and maneuvering to avoid them. The same technical solution is to be adapted for the data center satellites. Our assumption is that even though the solar panels are much larger than those on Starlink satellites, our assumption is that this issue can effectively be addressed, as it has been done on Starlink. The company is (and has been) well aware of this issue in space.

APPENDIX - TUTORIAL FOR HEAT TRANSFER MATHEMATICS

We are going to try to provide a (reasonably easy-to-follow) tutorial in basic heat transfer for you in order to answer the cooling question.

First, some basics:

- In basic thermodynamics heat flows from hot to cold.
- Objects transfer heat through three primary mechanisms:
 - **Conduction**, where thermal energy is transferred via direct contact of a hot object to a cooler one through molecular collisions, without bulk movement of the materials themselves. Think of a pan heating up when placed directly on an electric stove for example.
 - **Convection**, where thermal energy is transferred through the bulk movement of fluids, where a cold fluid can remove heat from a hot object. Think of your car's radiator circulating coolant to keep your engine cool.
 - **Thermal radiation**, where energy is directly emitted from a hot object to its surroundings via electromagnetic waves. Think of putting your hand close to a hot light bulb and feeling the heat. Unlike the other two, radiation requires no medium to travel through, it is just as effective in a vacuum.
- **Objects in the vacuum of space lose energy only through radiation** as there is no physical contact with anything and no fluid surrounding the object to remove heat.
- An object emits energy via radiation proportional to the FOURTH power of its own temperature minus the 4th (known as the Stephen-Boltzmann Law), or

$$E = \sigma T^4$$

which holds for a perfect blackbody.

- The proportionality constant is sigma, the Boltzmann constant, equal to $5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$.
- The net energy transfer from a radiating body is proportional to the fourth power of the temperature of the body (in K) minus the 4th power of the temperature of the surroundings it is radiating to, or

$$E_{net} = \sigma(T^4 - T_{surroundings}^4)$$

though in this case given the temperature of deep space is ~3 kelvin we will assume the temperature of the surroundings is effectively zero, suggesting we can use the Stephen-Boltzmann Law on its own.

- "Perfect" energy emission is known as "blackbody" radiation, and perfectly follows the Stephen-Boltzmann law. Not all objects emit perfectly however. Objects can have an emissivity (epsilon) between zero and 1, in other words

$$E = \epsilon \sigma T^4$$

- An object cooler than its surroundings will experience a rise in temperature (and corresponding increase in emitted energy as temp goes up) until it is emitting as much energy as it is receiving. At this point it is in "equilibrium" and its temperature will stop rising.

- At equilibrium all energies entering and leaving an object are in balance. We can use this state to calculate what we need to.

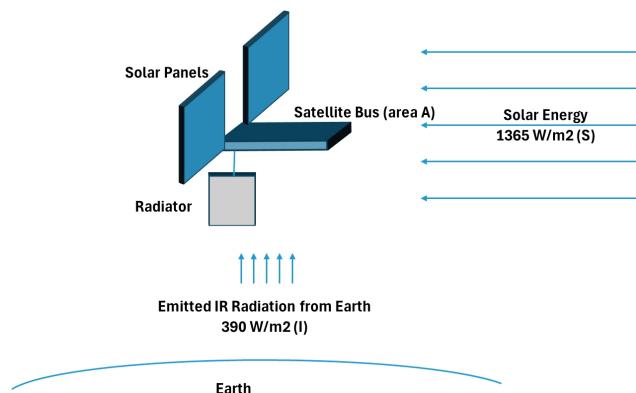
We are going to consider the orbital configuration shown in Exhibit 6. We assume a flat plate-like satellite bus for simplicity. In this case the satellite bus is oriented edge-on to the sun's rays; we will for simplicity assume that the area of the bus hence absorbs no solar energy, and that the edge of the bus that is exposed to solar radiation is too thin to capture much (hence we will ignore it). We will assume that the solar panels are oriented directly normal to the sun. This configuration would be representative of what is known as a sun-synchronous dawn/dusk orbit, where the satellite follows the day/night terminator and hence is always in the sun (Exhibit 7).

However, the sun is not the only source of energy impinging on the satellite. The earth itself emits, both via reflected sunlight (often up to a third of the solar energy hitting the planet) as well as through infrared emission. In this configuration there is no reflected sunlight, however the bottom of the satellite bus is exposed to earth-sourced IR radiation. The magnitude of this varies with time of day, seasons, position, and other things. However treating the earth as a perfect blackbody radiating at 15 degrees C would produce a flux of ~390 W/m², which we will use here.

Note that the solar cells do not turn all incident solar energy into power. But, the excess will be radiated away. We will assume the panels are at equilibrium and that this excess heat does not impinge on the satellite body (and hence will not have to be removed from it).

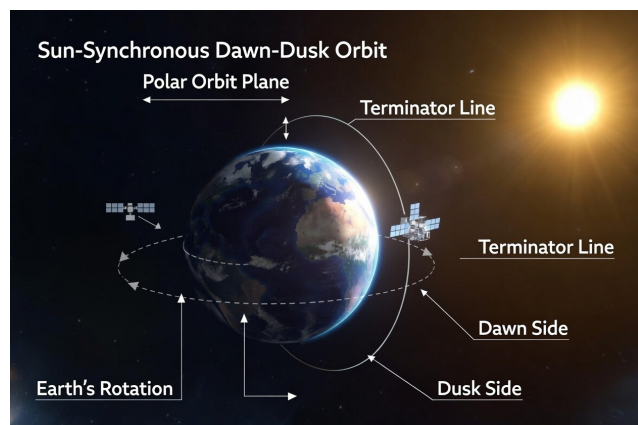
There are many other configurations that the satellite could be in. The sun could be shining directly down on it for example, or it could be positioned at an angle. The below configuration however likely represents a best-case scenario, and we will use there (the math is easier, and we are looking for proof on concept rather than rigorous analysis).

EXHIBIT 6: A datacenter satellite in orbit



Source: Bernstein estimates and analysis

EXHIBIT 7: A sun-synchronous dawn-dusk orbit hugs the terminator line, keeping the satellite in the sun



Source: Grok

We now set up our energy balance. Remember that the satellite will have to emit all energy it receives, plus all energy produced on board. In other words,

$$\text{Energy Emitted} = \text{Energy Absorbed} + \text{Energy Produced}$$

The energy absorbed is the sum of the solar energy hitting the satellite bus (zero in this configuration) plus the energy absorbed from the earth (or $I \cdot A$, where $I = 390 \text{ W/m}^2$, and A is the area of the satellite bus, i.e. what we are trying to solve for).

The energy produced by the panels is P (we will range this from 10 kW to 100 kW).

The energy emitted will come from both the top and the bottom of the satellite bus. Each side will radiate according to the Stephann-Boltzman law as

$$E = 2\epsilon\sigma T^4$$

(note the factor of 2 given the satellite can radiate from both sides of the bus). A is the area of the satellite that we are trying to find. Epsilon is the emissivity of the surface (we will take it as 0.85, not quite a perfect black body but close). Sigma is the Boltzman constant. And T is the final cooled temperature of the satellite (we will range this from 70 degrees C to 100 degrees C, or more appropriately 343 to 373 degrees Kelvin).

Setting up our balance, we find:

$$2A\epsilon\sigma T^4 = P + IA$$

Solving for A (the area needed for our satellite), we find:

$$A = \frac{P}{2T^4\epsilon\sigma - I}$$

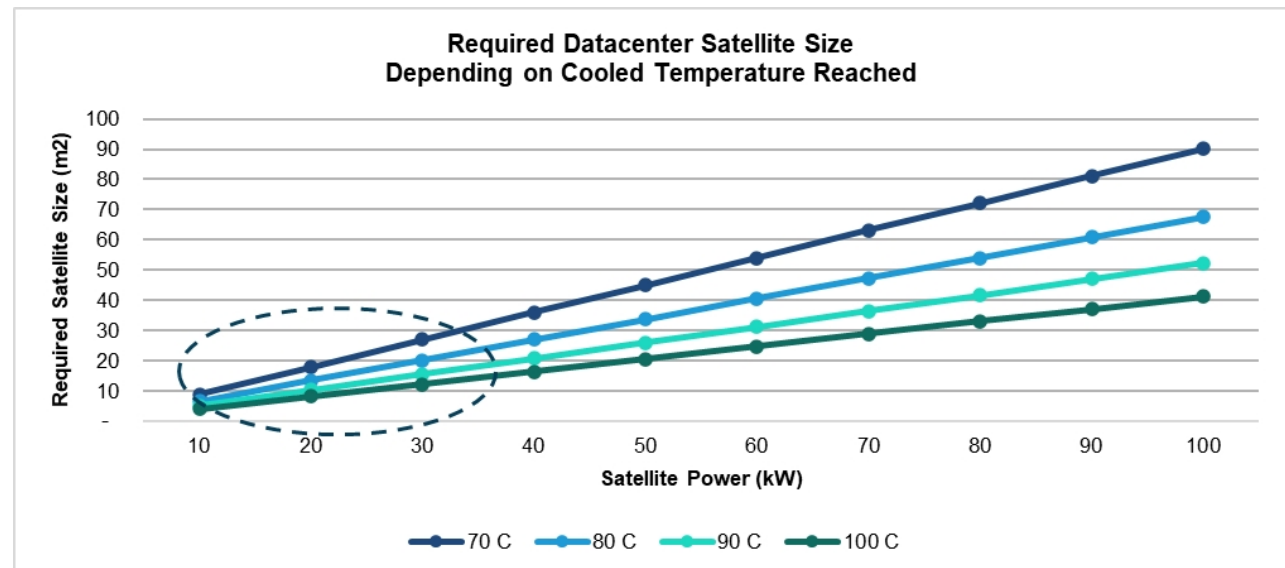
As an example, a 20 kW satellite in this configuration, cooled to 80 degrees C (353 K), would require an area of:

$$A = \frac{20,000 \text{ W}}{2(353 \text{ K})^4(0.85)(5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4) - 390 \text{ W/m}^2} = 18 \text{ m}^2$$

which does not seem unreasonable.

Ranging the math, we find that “launchable” satellites of ~30m² or less could probably be managed with up to ~30 kW of onboard compute power supplied by solar cells (Exhibit 8). Note that this is on the order of what the existing Starlink satellites look like in terms of power usage, and they are cooled radiatively just fine without the need for additional on-board radiators.

EXHIBIT 8: “Reasonably-sized” (i.e. launchable) satellites of up to ~30 kW could probably be cooled in orbit without additional radiators



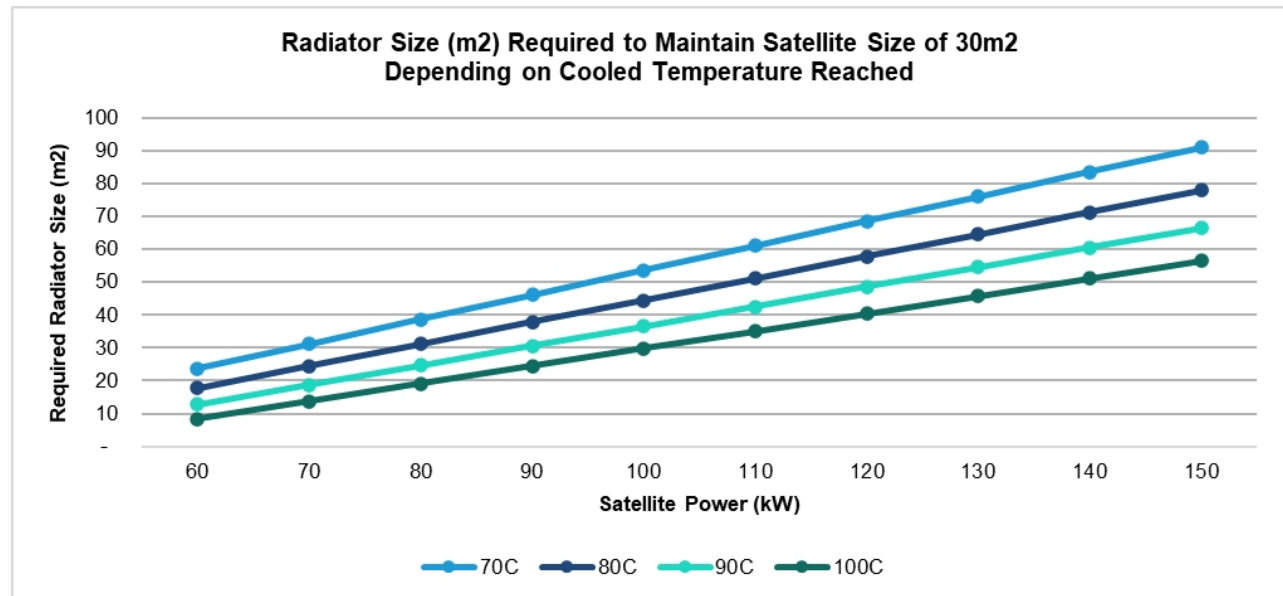
Source: Bernstein estimates and analysis

However, it seems likely that the increased power needs of a datacenter satellite (>100 kW) will not be satisfied by radiative cooling from the bus itself (the satellite would probably have to be too big to launch). But we are not limited by the satellite body itself as an emitter. Rather, we can add a radiator to the satellite to increase the area available to radiate energy away (and hence lower the required size of the satellite itself). In other words, we can fix the size of the satellite (at, say, 30 m²), and calculate

how much energy it could radiate on its own relative to the energy coming in. If it is not enough, we add a radiator, radiating away energy from both sides. The amount of the energy mismatch will tell us how big of a radiator we need. Note that we will orient the radiator perpendicular to the solar flux, and will assume it picks up no radiated IR from the earth (untrue in practice but easier to math).

In fact, this strategy enables us to get much more compute onto the satellite. The math suggests that we could keep a 100 kW satellite at a 30 m² size by including a radiator 30-50 m² in size depending on temperature, which seems doable (Exhibit 9).

EXHIBIT 9: **Adding a radiator would allow 100+ kW launchable satellites**



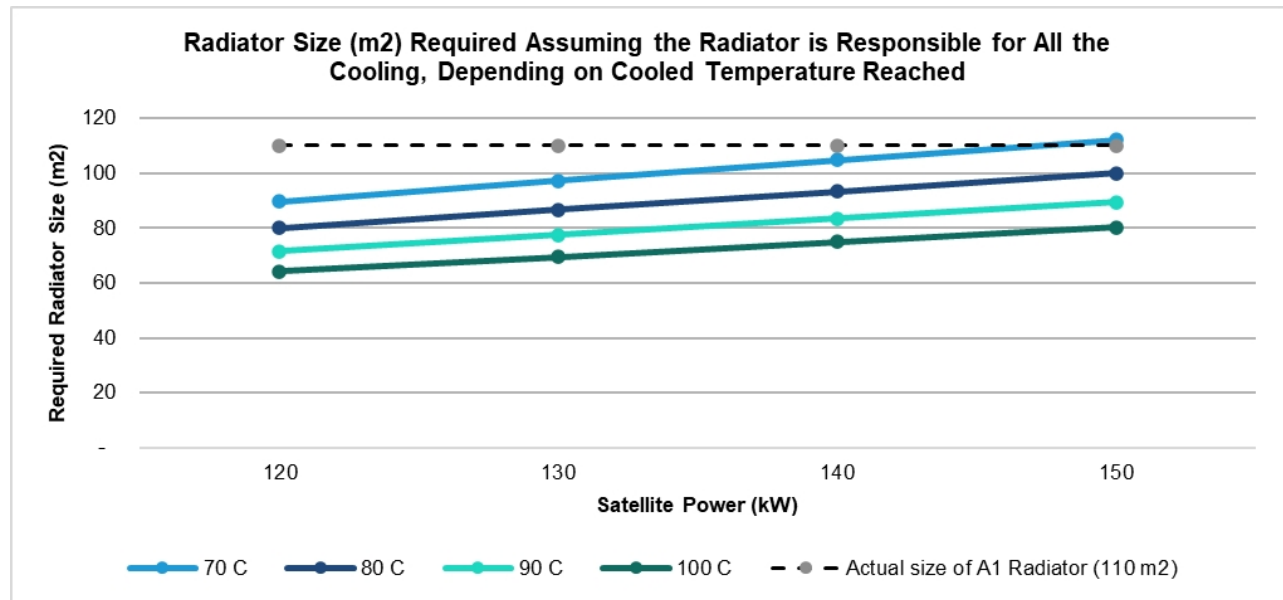
Source: Bernstein estimates and analysis

We note that SpaceX has in fact already outlined the architecture of their first-generation AI1 datacenter satellite, with an average power of 120 kW and a peak power output of 150 kW. At 250 W/m² for the solar panels this implies a panel area of ~600 m² (2 wings of ~300m² each) with a radiator of 110 m² (Exhibit 10).

Assuming that 100% of the heat removal from this satellite is done via the radiator (i.e. ignoring radiative cooling from the satellite body which is small in comparison) our math suggests a radiator size requirement very consistent with what the company is proposing, giving us confidence that our calculations are at least in the ballpark.

Long story short, we believe cooling 100+ kW datacenter satellites in orbit via radiative cooling is completely viable.

EXHIBIT 10: Our math suggests a radiator size requirement very consistent with what the company is proposing for their A1 satellite



Source: Company reports, Bernstein analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 11: SpaceX Annual Income Statement

SpaceX Annual Income Statement (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Space	3,796	4,086	4,794	7,560	7,513	7,224	7,930	9,167
Connectivity	7,599	11,387	16,941	27,727	53,120	94,679	140,998	205,908
AI	2,620	3,201	18,501	49,703	103,091	131,330	158,382	339,177
Total Revenues	\$ 14,015	\$ 18,674	\$ 40,236	\$ 84,990	\$ 163,725	\$ 233,233	\$ 307,310	\$ 554,251
Cost of revenue	(7,996)	(9,451)	(15,394)	(29,242)	(55,457)	(77,786)	(95,851)	(158,712)
Research & Development	(3,464)	(8,643)	(20,085)	(37,197)	(27,895)	(37,507)	(43,052)	(86,909)
Selling, General, and administrative	(1,813)	(2,644)	(3,534)	(5,010)	(9,008)	(12,411)	(16,157)	(25,705)
Restructuring charges	(213)	(487)	11	-	-	-	-	-
Impairment	(63)	(38)	-	-	-	-	-	-
Total cost and expenses	-13,549	-21,263	-39,002	-71,448	-92,359	-127,704	-155,060	-271,325
Space	21	(657)	(2,047)	1,262	896	236	503	874
Connectivity	2,006	4,423	6,517	12,164	24,626	44,437	67,304	102,316
AI	(1,561)	(6,355)	(3,236)	115	45,843	60,856	84,443	179,736
Operating income	\$ 466	\$ (2,589)	\$ 1,234	\$ 13,542	\$ 71,365	\$ 105,529	\$ 152,250	\$ 282,926
Interest income (expense)	(1,209)	(1,453)	(1,576)	(3,257)	(6,198)	(12,108)	(17,954)	(19,976)
Other income (expense), net	985	-177	-1876	0	0	0	0	0
Income (loss) before income taxes	\$ 242	\$ (4,219)	\$ (2,218)	\$ 10,285	\$ 65,167	\$ 93,421	\$ 134,296	\$ 262,950
Provisions for (benefit from) income taxes	549	(718)	(59)	(99)	14,202	(2,414)	(4,828)	(39,563)
Net income	\$ 791	\$ (4,937)	\$ (2,278)	\$ 10,186	\$ 79,369	\$ 91,006	\$ 129,467	\$ 223,387
Shares outstanding (mn)								
Basic			11,789	11,789	11,789	11,789	11,789	11,789
Diluted			12,344	12,344	12,344	12,344	12,344	12,344
EPS (\$)								
Basic EPS			-0.19	0.86	6.73	7.72	10.98	18.95
Diluted EPS			-0.18	0.83	6.43	7.37	10.49	18.10
Adjusted EBITDA	\$ 5,350	\$ 6,584	\$ 17,131	\$ 43,447	\$ 112,068	\$ 159,070	\$ 216,453	\$ 403,516
Space	1,154	653	-544	2,817	3,122	2,939	3,768	4,556
Connectivity	3,849	7,168	10,308	17,233	30,947	53,498	78,205	114,737
AI	347	-1,237	7,367	23,397	77,999	102,634	134,480	284,224

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 12: **SpaceX Quarterly Income Statement**

SpaceX Quarterly Income Statement (\$m)											
Fiscal Year Ending December 31											
	2024					2025					2026E
	12 mo	Q1	Q2	Q3	Q4	12 mo	Q1	Q2	Q3	Q4	12 mo
Space	3,796	865	746	1,666	809	4,086	619	855	1,538	1,782	4,794
Connectivity	7,599	2,475	2,588	3,096	3,228	11,387	3,257	3,763	4,421	5,500	16,941
AI	2,620	727	737	814	923	3,201	818	1,928	5,640	10,115	18,501
Total Revenues	\$ 14,015	\$ 4,067	\$ 4,071	\$ 5,576	\$ 4,960	\$ 18,674	\$ 4,694	\$ 6,546	\$ 11,599	\$ 17,397	\$ 40,236
Cost of revenue	(7,996)	(1,962)	(2,282)	(2,706)	(2,501)	(9,451)	(2,388)	(2,981)	(4,171)	(5,854)	(15,394)
Research & Development	(3,464)	(1,557)	(1,958)	(2,291)	(2,837)	(8,643)	(3,514)	(4,434)	(4,785)	(7,352)	(20,085)
Selling, General, and administrative	(1,813)	(493)	(606)	(575)	(970)	(2,644)	(746)	(861)	(896)	(1,032)	(3,534)
Restructuring charges	(213)	(4)	(189)	(278)	(16)	(487)	11	-	-	-	11
Impairment	(63)	(24)	(6)	1	(9)	(38)	-	-	-	-	-
Total cost and expenses	-13,549	-4,040	-5,041	-5,849	-6,333	-21,263	-6,637	-8,275	-9,851	-14,239	-39,002
Space	21	-70	-370	258	-475	-657	-662	-756.2535	-319.3394	-309.4451	-2047.038
Connectivity	2006	1033	924	1189	1277	4423	1188	1300.0126	1759.9837	2269.1207	6517.117
AI	(1,561)	(936)	(1,524)	(1,720)	(2,175)	(6,355)	(2,469)	(2,273)	307	1,199	(3,236)
Operating income (loss) from operations	\$ 466	\$ 27	\$ (970)	\$ (273)	\$ (1,373)	\$ (2,589)	\$ (1,943)	\$ (1,729)	\$ 1,747	\$ 3,159	\$ 1,234
Interest income (expense)	(1,209)	(330)	(312)	(359)	(452)	(1,453)	(451)	(633)	(223)	(269)	(1,576)
Other income (expense), net	985	(211)	413	140	(519)	(177)	(1,876)	-	-	-	(1,876)
Income (loss) before income taxes	\$ 242	\$ (514)	\$ (869)	\$ (492)	\$ (2,344)	\$ (4,219)	\$ (4,270)	\$ (2,362)	\$ 1,524	\$ 2,890	\$ (2,218)
Provisions for (benefit from) income taxes	549	(14)	(138)	(562)	(4)	(718)	(6)	-	-	(53)	(59)
Net income	\$ 791	\$ (528)	\$ (1,007)	\$ (1,054)	\$ (2,348)	\$ (4,937)	\$ (4,276)	\$ (2,362)	\$ 1,524	\$ 2,836	\$ (2,278)
Shares outstanding (mn)											
Basic							4,496	5,950	11,789	11,789	11,789
Diluted							5,068	6,507	12,344	12,344	12,344
EPS (\$)											
Basic EPS							-\$0.95	-\$0.40	\$0.13	\$0.24	-\$0.19
Diluted EPS							-\$0.84	-\$0.36	\$0.12	\$0.23	-\$0.18
Adjusted EBITDA	5,350	1,730	1,212	2,168	1,474	6,584	1,127	2,129	6,128	7,748	17,131
Space	1,154	224	-94	672	-149	653	-351	-379	80	106	-544
Connectivity	3,849	1,618	1,583	1,921	2,046	7,168	2,087	2,223	2,719	3,279	10,308
AI	347	-112	-277	-425	-423	-1,237	-609	285	3,329	4,362	7,367

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 13: **SpaceX Annual Cash Flow**

SpaceX Annual Cash Flow (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Net income (loss)	791	(4,937)	(2,278)	10,186	79,369	91,006	129,467	223,387
Depreciation & Amortization	3,824	6,701	12,993	26,621	36,923	49,352	59,556	115,514
Stock Based Compensation	784	1,947	2,915	3,284	3,779	4,189	4,647	5,077
Impairment & Other Charges	63	38	-	-	-	-	-	-
Non-Cash Interest Income and Other	(1,359)	947	998	(703)	(1,031)	(330)	(1,340)	(1,179)
Accounts Receivable	(347)	(543)	(1,838)	(1,240)	(7,320)	(7,385)	(7,076)	(7,881)
Inventory	(309)	(413)	(1,477)	(802)	805	245	(6)	259
Prepaid Expenses & Other	(328)	(673)	(1)	(70)	216	(77)	(66)	(143)
Accounts Payable	472	709	(4,424)	4,273	3,188	7,228	7,066	7,555
Deferred Revenue	1,876	1,929	11,382	809	(3,580)	1,663	6,497	(3,114)
Other Liabilities	309	1,080	4,616	(102)	(1,256)	311	251	420
Net cash provided by operating activities	\$ 5,776	\$ 6,785	\$ 22,887	\$ 42,257	\$ 111,095	\$ 146,203	\$ 198,996	\$ 339,894
Purchase of PPE	(11,163)	(20,619)	(47,643)	(144,226)	(179,933)	(201,296)	(301,139)	(370,695)
Other investing activities, net	367	1,044	(7,812)	-	-	-	-	-
Net cash used in investing activities	\$ (10,796)	\$ (19,575)	\$ (55,455)	\$ (144,226)	\$ (179,933)	\$ (201,296)	\$ (301,139)	\$ (370,695)
Debt Issuance / (Paydown)	(77)	9,131	7,912	77,668	121,525	96,008	45,953	(20,675)
Finance Lease Payments	(154)	(295)	(373)	(470)	(393)	-	-	-
Equity Issuance / (Repurchases) **	12,061	17,514	62,893	(1,653)	(2,432)	(2,553)	(2,123)	(2,781)
Net cash provided by financing activities	\$ 11,830	\$ 26,350	\$ 70,431	\$ 75,545	\$ 118,699	\$ 93,454	\$ 43,830	\$ (23,456)
Effect of exchange rate changes on cash and cash equivalents	1	63	36	-	-	-	-	-
Net change in cash and cash equivalents and restricted cash	\$ 6,811	\$ 13,623	\$ 37,900	\$ (26,425)	\$ 49,861	\$ 38,361	\$ (58,313)	\$ (54,258)
Cash and cash equivalents and restricted cash, beginning year	\$ 4,690	\$ 11,501	\$ 25,124	\$ 63,024	\$ 36,599	\$ 86,461	\$ 124,822	\$ 66,509
Cash and cash equivalents and restricted cash, end year	\$ 11,501	\$ 25,124	\$ 63,024	\$ 36,599	\$ 86,461	\$ 124,822	\$ 66,509	\$ 12,252
FCF	(5,387)	(13,834)	(24,756)	(101,970)	(68,838)	(55,093)	(102,143)	(30,801)
Conversion to net income %	-681%	280%	1087%	-1001%	-87%	-61%	-79%	-14%
Working Capital	(12,177)	(20,272)	(28,530)	(31,398)	(23,452)	(25,438)	(32,104)	(29,199)

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 14: **SpaceX Annual Balance Sheet**

SpaceX Annual Balance Sheet (\$m)								
Fiscal Year Ending December 31	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Cash and cash equivalents	11,385	24,747	62,268	35,843	85,705	124,066	65,754	11,496
Marketable securities	800	0	7,823	7,823	7,823	7,823	7,823	7,823
Account receivables (net allowance credit loss)	1,052	1,579	3,417	4,657	11,977	19,362	26,437	34,318
Inventory	2,003	2,416	3,893	4,695	3,890	3,644	3,650	3,392
Prepaid expenses and other current assets	868	2,210	2,211	2,280	2,065	2,142	2,208	2,351
Total current assets	16,108	30,952	79,611	55,298	111,458	157,036	105,871	59,379
Property, plant, and equipment, net	21,147	42,602	77,251	194,856	337,865	489,809	731,392	986,573
Finance lease right-of-use assets	1,686	1,260	892	422	0	0	0	0
Intangible assets, net	2,211	1,548	0	0	0	0	0	0
Digital assets	1,749	1,637	0	0	0	0	0	0
Goodwill	11,129	11,809	0	0	0	0	0	0
Other assets	2,336	2,130	21,354	34,157	22,056	32,583	44,210	55,965
Total non current assets	40,258	60,986	99,497	229,435	359,921	522,392	775,602	1,042,538
Total assets	56,366	91,938	179,108	284,733	471,380	679,428	881,473	1,101,917
Accounts payable	4,413	11,792	7,368	11,641	14,829	22,057	29,123	36,678
Deferred revenue, current	10,179	12,116	23,498	24,307	20,728	22,390	28,887	25,773
Accrued expenses and other liabilities	1,508	2,569	7,185	7,082	5,827	6,138	6,389	6,809
Total current liabilities	16,100	26,477	38,051	43,030	41,383	50,585	64,399	69,259
Debt	12,262	21,659	33,800	111,468	232,993	329,001	374,954	354,279
Finance Lease Liabilities	1,531	1,237	863	393	0	0	0	0
Deferred Tax Liability / (Asset)	-725	-113	8	84	-14,305	-5,136	4,032	13,201
Other Long-Term Liabilities	1,394	1,353	1,293	1,293	1,293	1,293	1,293	1,293
Total non-current liabilities	14,462	24,136	35,964	113,238	219,981	325,158	380,280	368,773
Total liabilities	30,562	50,613	74,015	156,268	261,365	375,743	444,679	438,033
Convertible Preferred Stock	20,941	38,752	0	0	0	0	0	0
Additional Paid-in Capital	35,868	37,710	142,509	155,693	157,874	160,538	164,180	167,883
Retained Earnings	-31,005	-35,137	-37,415	-27,229	52,141	143,147	272,615	496,002
Total shareholders' equity	25,804	41,325	105,094	128,465	210,015	303,685	436,795	663,884
Total liabilities and equity	56,366	91,938	179,109	284,733	471,379	679,428	881,473	1,101,917
Total Debt	13,793	22,896	34,663	111,861	232,993	329,001	374,954	354,279
Net debt	2,408	-1,851	-27,605	76,018	147,288	204,935	309,201	342,783

Source: Company reports, Bernstein analysis and estimates

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	30 Jul 2026	Price Target	TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
			Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
SPCX (SpaceX)	O	USD	112.20	239.00	NA	USD		(0.18)	0.83		(608.1)	136.0
SPX			7,437.63									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY**Space Exploration Technologies Corporation**

We base our valuation approach on 12-month forward EV/EBITDA applied to 2031 EBITDA. We believe it is necessary to use a long-term period (i.e., 2031) because the cornerstone of the value is in the orbital data centers. Applying a multiple to, say, 2028 EBITDA would not capture any of that business - i.e., it would simply be space launch, terrestrial compute and near-term Starlink. We use a sum of the parts methodology to give an EV/EBITDA valuation in 2030 (based on 2031 values). We then discount that valuation backward for each segment to mid-2027 in order to provide a one year target price of \$239.

RISKS**Space Exploration Technologies Corporation**

Downside Risks to our Outperform rating and \$239 price target include the ability to dramatically growth Starship launch capacity over time, which depends on full reusability. SpaceX must be able to operationalize its orbital data centers to deliver on the valuation. Semiconductor availability could also be a constraint, as could other potential supply chain issues. There are also risks associated with potential launch incidents or regulatory restrictions. Should compute demand be more limited than most expectations, that would form another risk.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors

on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
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- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

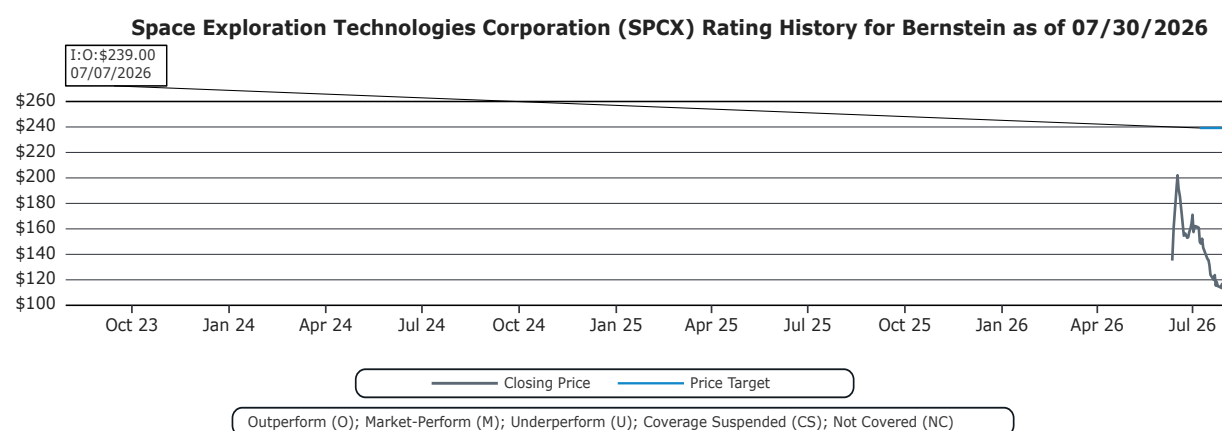
DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

CONFLICTS OF INTEREST

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Stacy A. Rasgon maintains long positions in various crypto currencies.

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Bernstein and/or affiliates have received compensation for non-investment banking securities-related products or services in the previous twelve months from the following clients: Space Exploration Technologies Corporation.

Bernstein and/or affiliates expect to receive or intend to seek compensation for investment banking services in the next three months from Space Exploration Technologies Corporation.

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Bernstein and/or affiliates had an investment banking client relationship during the past twelve months with Space Exploration Technologies Corporation.

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